



Statistical Methods & Machine Learning 1

Class #5

John Kubale

September 24, 2024

Outline

- Review
- Multiple Linear Regression
 - Why we do it
 - DAGs
 - Multiple Linear Regression Estimation
- Interpretation of Regression Coefficients
- Inference about Regression Coefficients
- Analysis of Variance in Linear Regression
- Model Comparison
- Goodness of Model Fit



Review

“As a function of”

- HW₃ included question: “Analyze the crime rates (`crim`) as a function of median home values (`medv`) in a simple linear regression...”
 - Which is the dependent variable (i.e., your outcome)?
 - Which is the independent variable (i.e., your predictor)?

“As a function of”

- HW included question: “Analyze the crime rates (`crim`) as a function of median home values (`medv`) in a simple linear regression...”
 - Which is the dependent variable (i.e., your outcome)?
 - Which is the independent variable (i.e., your predictor)?
- `lm(crim ~ medv, Boston)`
- Put another way: $E(crim) = f(medv) \rightarrow$ here the “function” is the linear model

Training vs. test data

- Training data are the model used to fit (or train) your model
- Test data are data not used in fitting/training the model, but that the model is used on to make predictions or inferences (i.e., they're previously unseen)
- Each has their own error (i.e., training error and test error)
- Which do you think is most important for assessing model fit?

Training vs. test data

- Training data are the model used to fit (or train) your model
- Test data are data not used in fitting/training the model, but that the model is used on to make predictions or inferences
- Each has their own error (i.e., training error and test error)
- Which do you think is most important for assessing model fit?
 - Can you see any challenges in using it for assessing model fit?



Why do we sometimes include multiple
predictors?

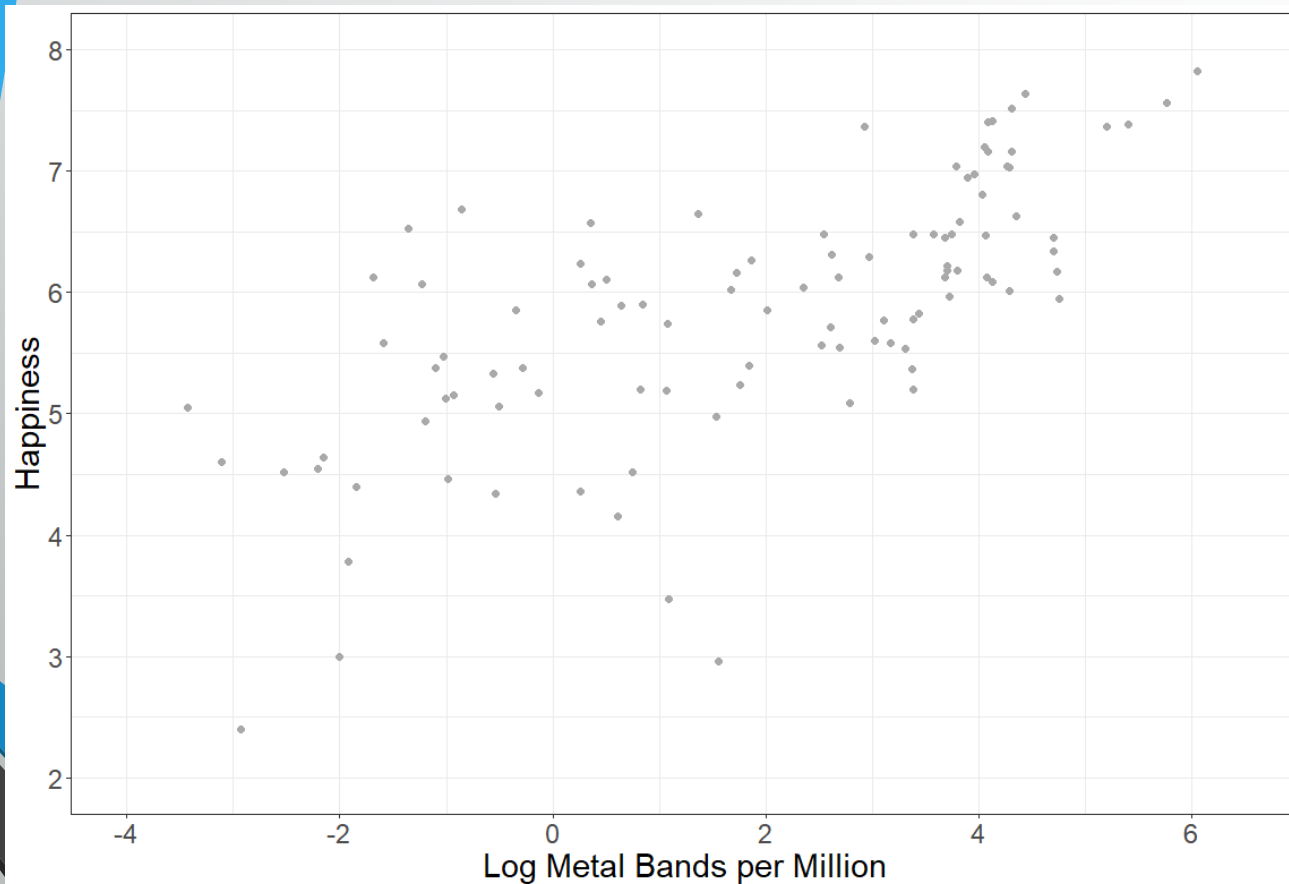
Why we sometimes include multiple predictors

- Allows us to test for a given association between a predictor/outcome while accounting for influence of other variables.
- Allows us to predict what will happen to an outcome when a predictor changes, while keeping other variables constant.

Why we sometimes include multiple predictors

- Allows us to test for a given association between a predictor/outcome while accounting for influence of other variables.
- Allows us to predict what will happen to an outcome when a predictor changes, while keeping other variables constant.
- Let's look at an example to see why this is useful in practice.

Heavy metal happiness



```
Call:
lm(formula = Happiness ~ log(bands_pm), data = metal_bands2)

Residuals:
    Min       1Q   Median       3Q      Max
-2.79612 -0.45529  0.04414  0.48408  1.67410

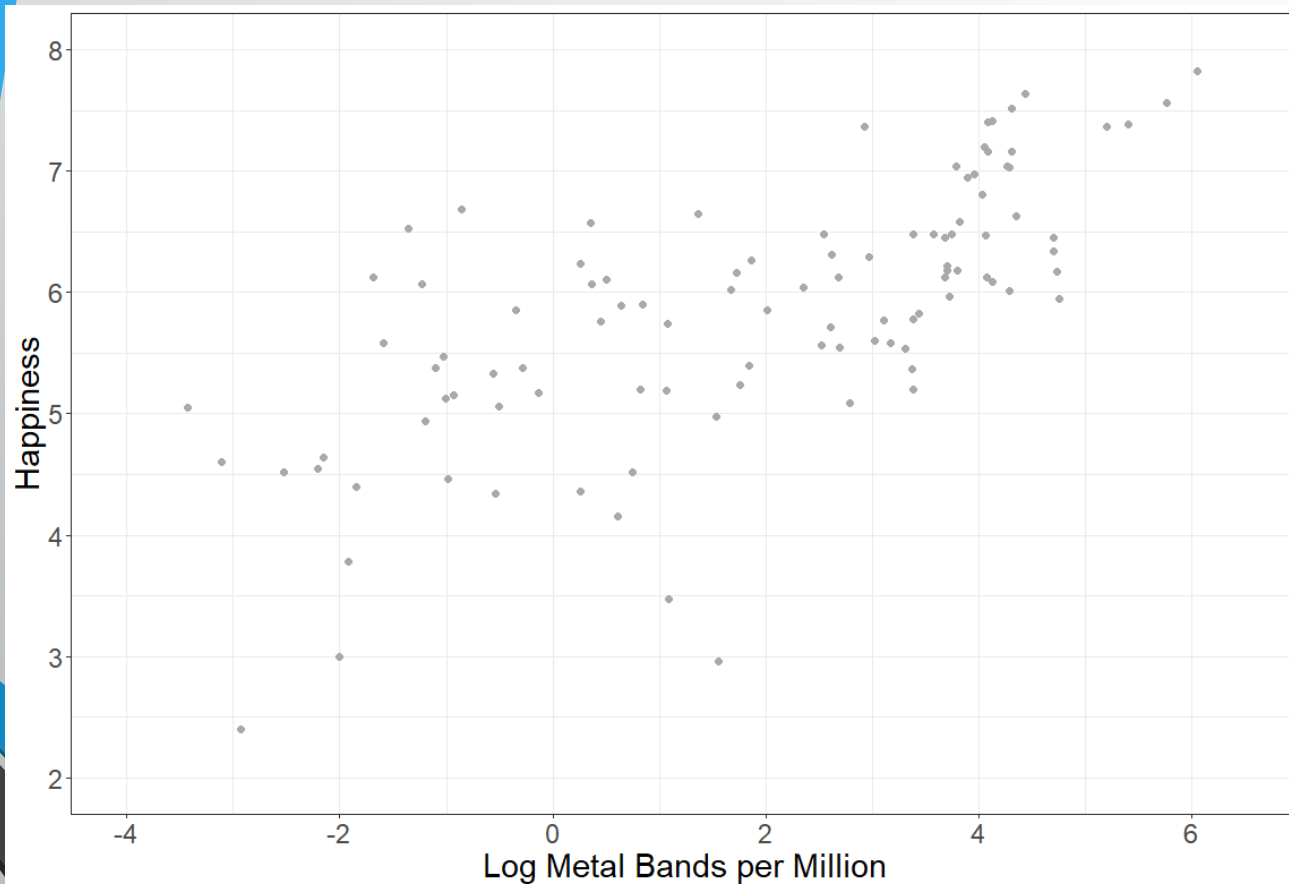
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.27412    0.09646   54.678  < 2e-16 ***
log(bands_pm)  0.30607    0.03216    9.518 1.04e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7652 on 101 degrees of freedom
(44 observations deleted due to missingness)
Multiple R-squared:  0.4728,    Adjusted R-squared:  0.4676
F-statistic: 90.58 on 1 and 101 DF,  p-value: 1.037e-15
```

Adapted from McElreath 2023

Heavy metal happiness

How would you interpret the slope for the predictor?



```
Call:
lm(formula = Happiness ~ log(bands_pm), data = metal_bands2)

Residuals:
    Min       1Q   Median       3Q      Max
-2.79612 -0.45529  0.04414  0.48408  1.67410

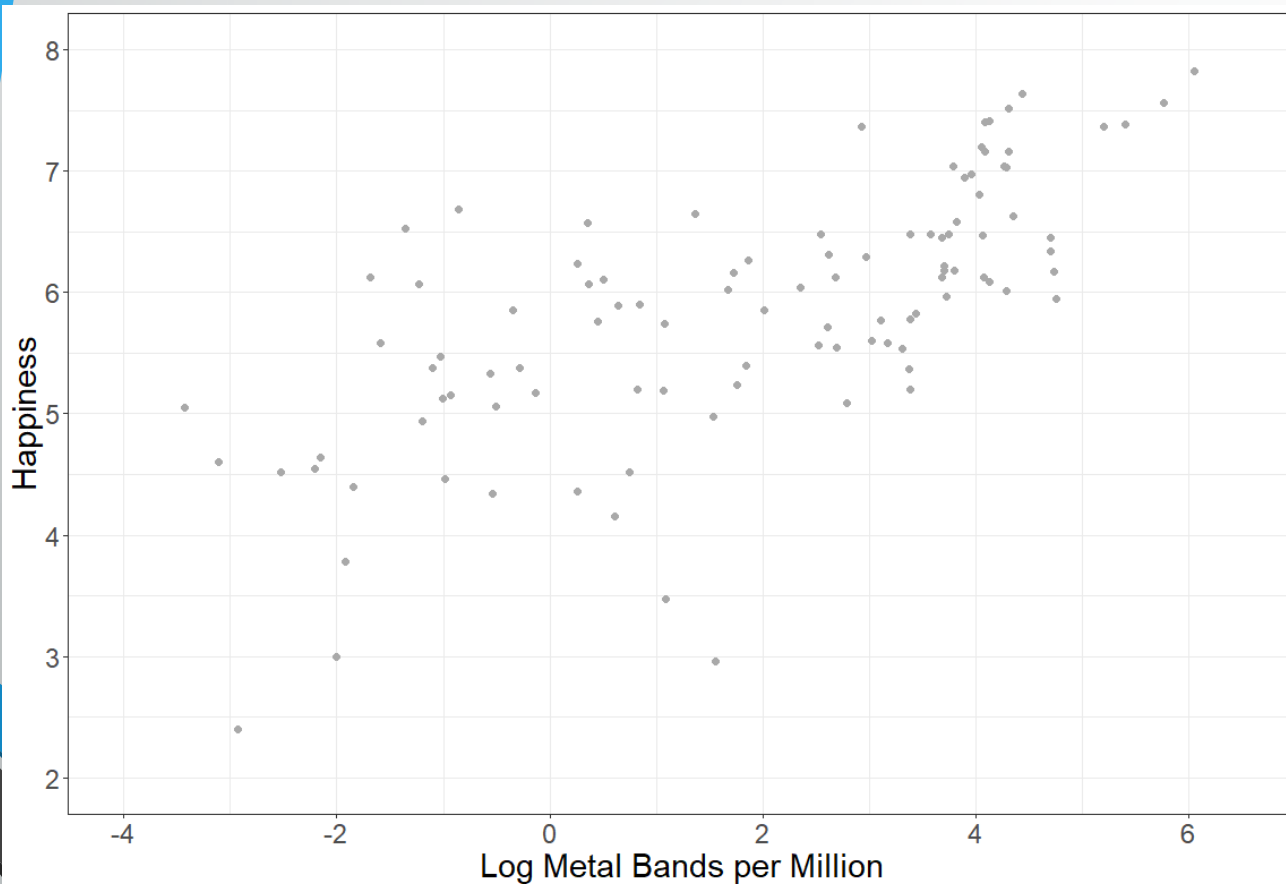
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.27412    0.09646   54.678  < 2e-16 ***
log(bands_pm)  0.30607    0.03216    9.518 1.04e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7652 on 101 degrees of freedom
(44 observations deleted due to missingness)
Multiple R-squared:  0.4728,    Adjusted R-squared:  0.4676
F-statistic: 90.58 on 1 and 101 DF,  p-value: 1.037e-15
```

Adapted from McElreath 2023

Heavy metal happiness

How would you interpret the slope for the predictor?



- A one-unit increase in log metal bands per million is associated with an increase of 0.3 in mean expected happiness score.
- A 10% increase in bands per million is associated with an increase in mean expected happiness of $0.30607 * \log(1.1) = 0.03$.

```
Call:
lm(formula = Happiness ~ log(bands_pm), data = metal_bands2)

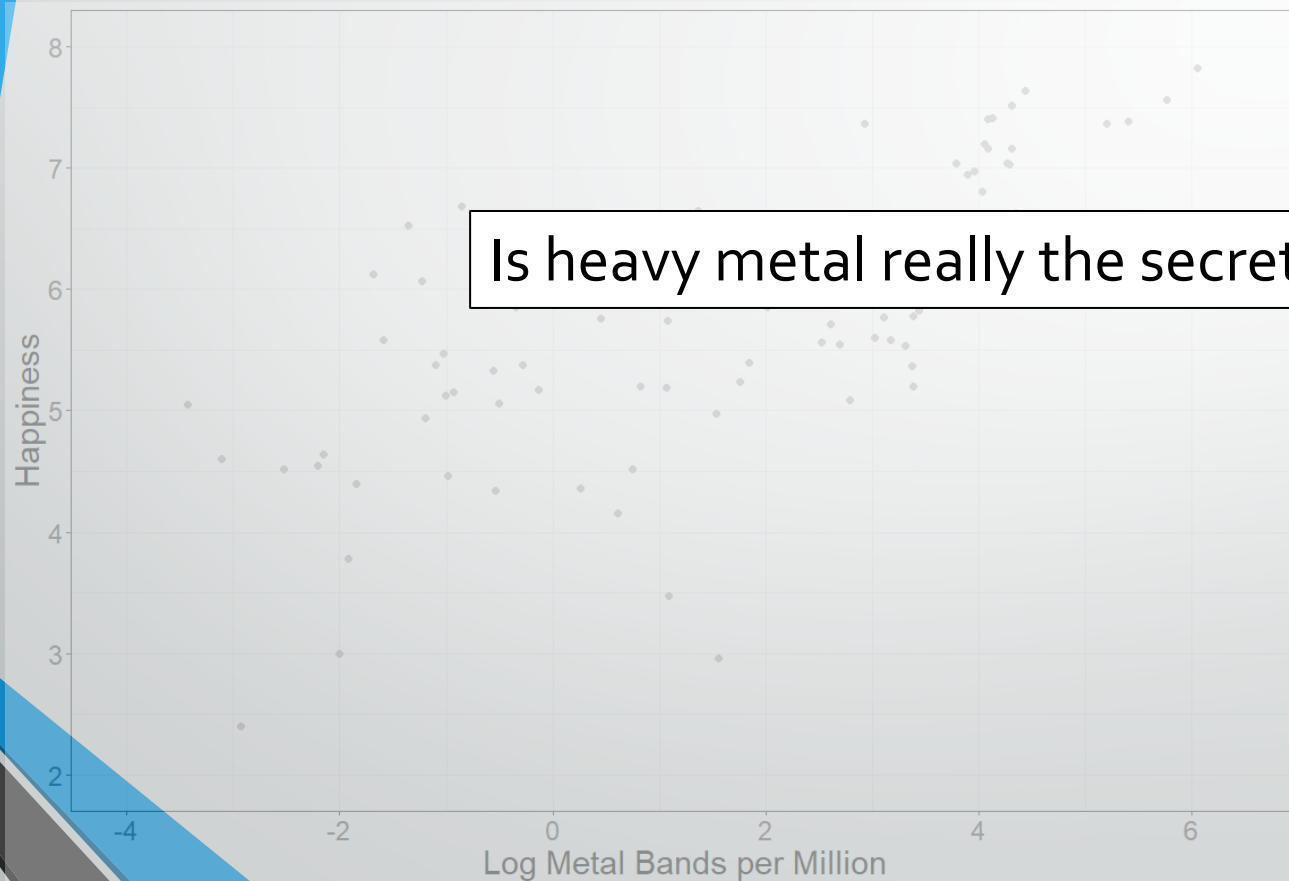
Residuals:
    Min       1Q   Median       3Q      Max
-2.79612 -0.45529  0.04414  0.48408  1.67410

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.27412    0.09646   54.678  < 2e-16 ***
log(bands_pm)  0.30607    0.03216    9.518 1.04e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7652 on 101 degrees of freedom
(44 observations deleted due to missingness)
Multiple R-squared:  0.4728,    Adjusted R-squared:  0.4676
F-statistic: 90.58 on 1 and 101 DF,  p-value: 1.037e-15
```

Adapted from McElreath 2023

Heavy metal happiness



```
Call:
lm(log(happiness) ~ log(bands_pm), data = metal_bands2)

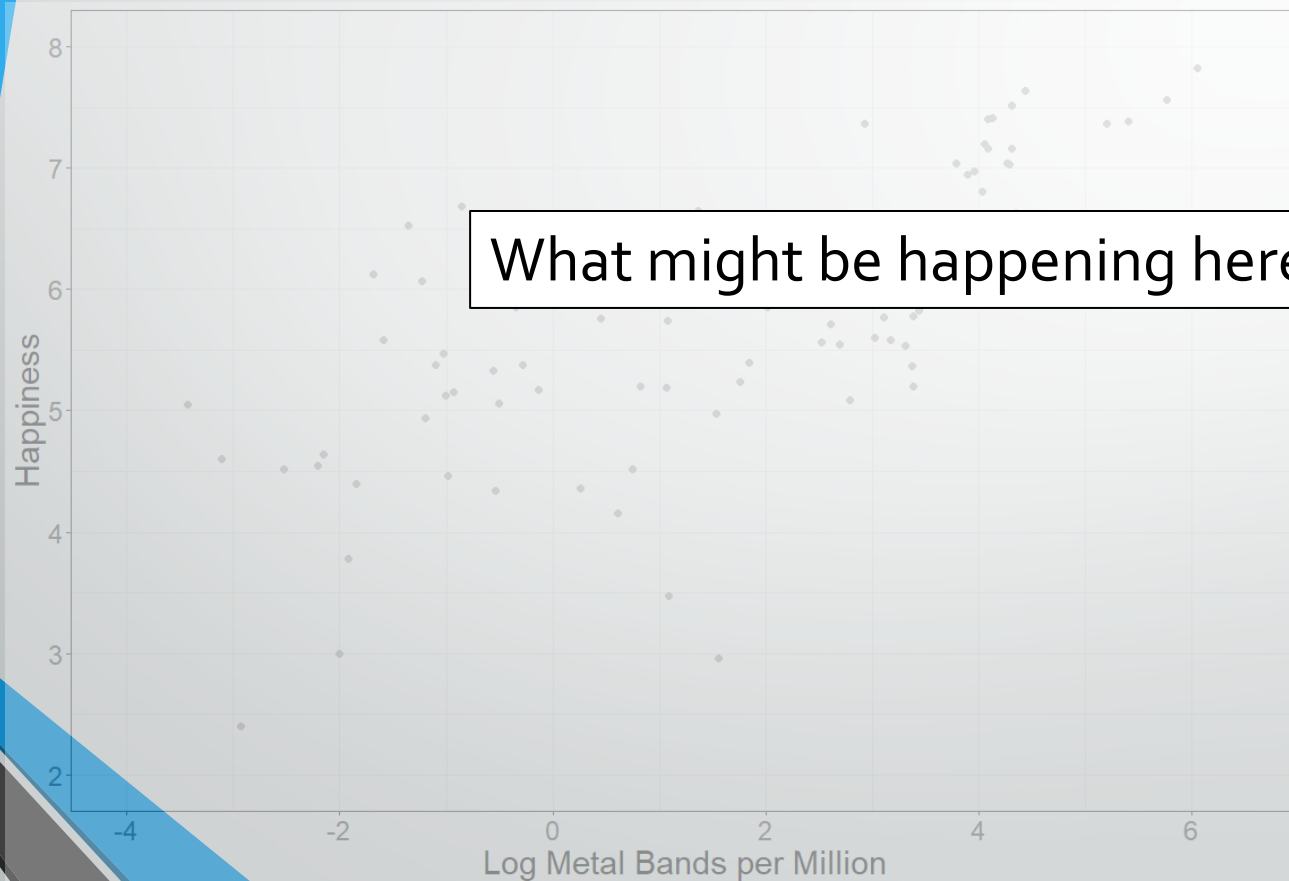
      Min       1Q   Median       3Q      Max 
-2.79612 -0.45529  0.04414  0.48408  1.67410 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    5.27412    0.09646   54.678  < 2e-16 ***
log(bands_pm)  0.30607    0.03216    9.518 1.04e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7652 on 101 degrees of freedom
(44 observations deleted due to missingness)
Multiple R-squared:  0.4728,    Adjusted R-squared:  0.4676 
F-statistic: 90.58 on 1 and 101 DF,  p-value: 1.037e-15
```

Adapted from McElreath 2023

Heavy metal happiness



```
Call:
lm(formula = Happiness ~ log(bands_pm), data = metal_bands2)

Residuals:
    Min       1Q   Median       3Q      Max
-2.79612 -0.45529  0.04414  0.48408  1.67410

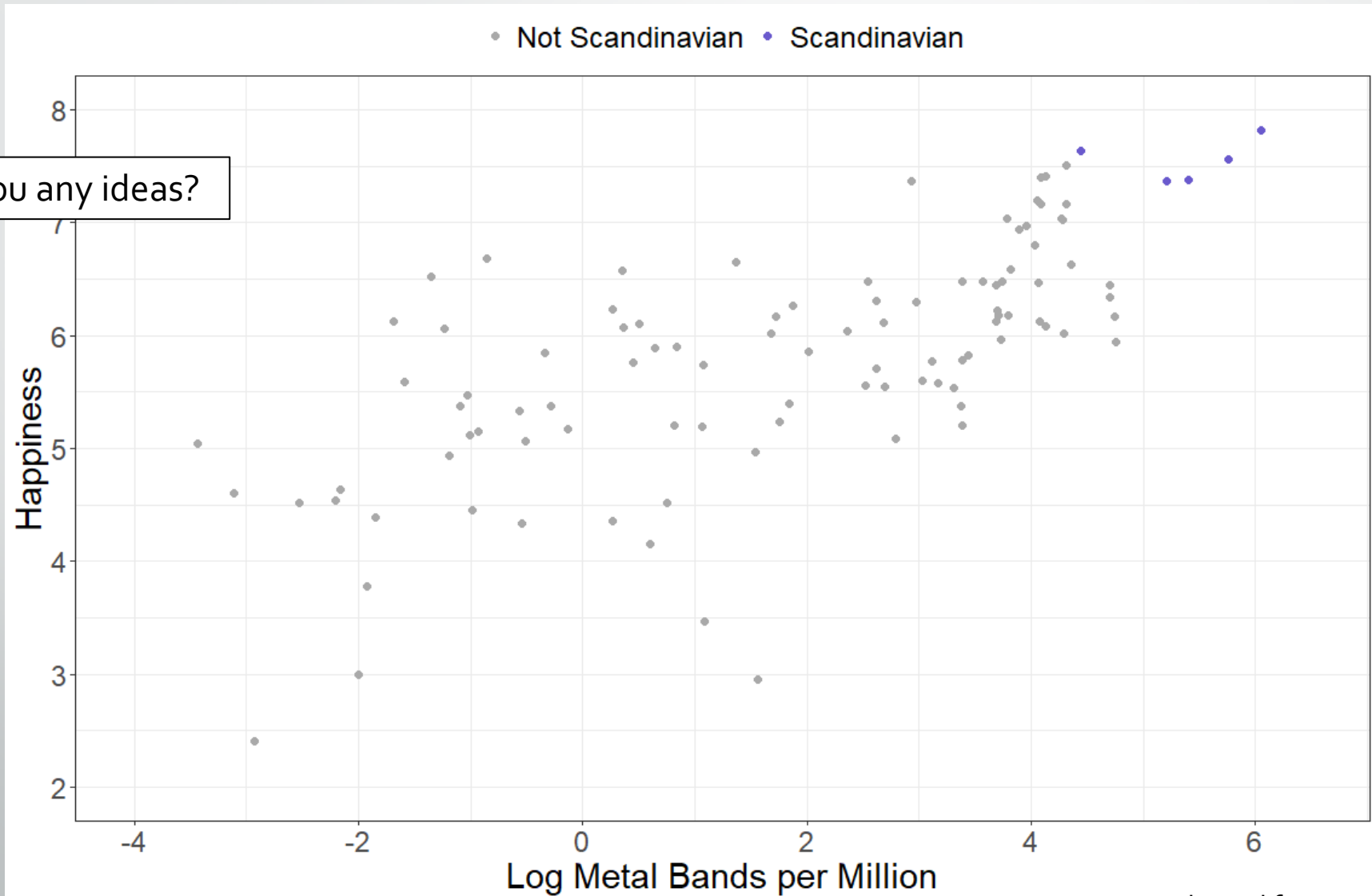
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.27412    0.09646   54.678  < 2e-16 ***
log(bands_pm)  0.30607    0.03216    9.518 1.04e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7652 on 101 degrees of freedom
(44 observations deleted due to missingness)
Multiple R-squared:  0.4728,    Adjusted R-squared:  0.4676
F-statistic: 90.58 on 1 and 101 DF,  p-value: 1.037e-15
```

Adapted from McElreath 2023

Heavy metal happiness

Does this give you any ideas?



Heavy metal happiness

- What happens if we add a measure of relative affluence like GDP per capita to the model?

```
call:
lm(formula = Happiness ~ log(bands_pm) + log(gdp_pc_2021), data = metal_bands2)

Residuals:
    Min       1Q   Median       3Q      Max
-2.37413 -0.24014  0.01006  0.34185  1.35327

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.73622    0.57267   1.286   0.2016
log(bands_pm)   0.08171    0.03775   2.165   0.0328 *
log(gdp_pc_2021) 0.53624    0.06708   7.994 2.34e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6006 on 100 degrees of freedom
(19 observations deleted due to missingness)
Multiple R-squared:  0.6784,    Adjusted R-squared:  0.6719
F-statistic: 105.5 on 2 and 100 DF,  p-value: < 2.2e-16
```

Heavy metal happiness

- What happens if we add a measure of relative affluence like GDP per capita from the World Bank to the model?

```
call:
lm(formula = Happiness ~ log(bands_pm), data = metal_bands2)

Residuals:
    Min       1Q   Median       3Q      Max
-2.79612 -0.45529  0.04414  0.48408  1.67410

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.27412    0.09646   54.678  < 2e-16 ***
log(bands_pm)  0.30607    0.03216   9.518 1.04e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7652 on 101 degrees of freedom
(44 observations deleted due to missingness)
Multiple R-squared:  0.4728,    Adjusted R-squared:  0.4676
F-statistic: 90.58 on 1 and 101 DF,  p-value: 1.037e-15
```

```
call:
lm(formula = Happiness ~ log(bands_pm) + log(gdp_pc_2021), data = metal_bands2)

Residuals:
    Min       1Q   Median       3Q      Max
-2.37413 -0.24014  0.01006  0.34185  1.35327

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.73622    0.57267   1.286  0.2016
log(bands_pm)  0.08171    0.03775   2.165  0.0328 *
log(gdp_pc_2021) 0.53624    0.06708   7.994 2.34e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6006 on 100 degrees of freedom
(19 observations deleted due to missingness)
Multiple R-squared:  0.6784,    Adjusted R-squared:  0.6719
F-statistic: 105.5 on 2 and 100 DF,  p-value: < 2.2e-16
```

Heavy metal happiness

- What happens if we add a measure of relative affluence like GDP per capita from the World Bank to the model?

```
call:
lm(formula = Happiness ~ log(bands_pm), data = metal_bands2)

Residuals:
    Min       1Q   Median       3Q      Max
-2.79612 -0.45529  0.04414  0.48408  1.67410

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.27412    0.09646   54.678 < 2e-16 ***
log(bands_pm)  0.30607    0.03216   9.518 1.04e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7652 on 101 degrees of freedom
(44 observations deleted due to missingness)
Multiple R-squared:  0.4728,    Adjusted R-squared:  0.4676
F-statistic: 90.58 on 1 and 101 DF,  p-value: 1.037e-15
```

A 10% increase of metal bands per million is associated with an increase in mean expected happiness of 0.03.

```
call:
lm(formula = Happiness ~ log(bands_pm) + log(gdp_pc_2021), data = metal_bands2)

Residuals:
    Min       1Q   Median       3Q      Max
-2.37413 -0.24014  0.01006  0.34185  1.35327

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.73622    0.57267   1.286  0.2016
log(bands_pm)  0.08171    0.03775   2.165  0.0328 *
log(gdp_pc_2021) 0.53624    0.06708   7.994 2.34e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6006 on 100 degrees of freedom
(19 observations deleted due to missingness)
Multiple R-squared:  0.6784,    Adjusted R-squared:  0.6719
F-statistic: 105.5 on 2 and 100 DF,  p-value: < 2.2e-16
```

A 10% increase of metal bands per million is associated with an increase in mean expected happiness of 0.008 after adjusting for GDP per capita.

Heavy metal happiness

- What happens if we add a measure of relative affluence like GDP per capita to the model?

```
call:
lm(formula = Happiness ~ log(bands_pm), data = metal_bands2)

Residuals:
    Min       1Q   Median       3Q      Max
-2.79612 -0.45529  0.04414  0.48408  1.60000

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.27412    0.09646  54.678  < 2e-16 ***
log(bands_pm)  0.30607    0.03216   9.518 1.04e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7652 on 101 degrees of freedom
(44 observations deleted due to missingness)
Multiple R-squared:  0.4728,    Adjusted R-squared:  0.4676
F-statistic: 90.58 on 1 and 101 DF,  p-value: 1.037e-15
```

A 10% increase of metal bands per million is associated with an increase in mean expected happiness of 0.03.

What do you think this means?

```
call:
lm(formula = Happiness ~ log(bands_pm) + log(gdp_pc_2021), data = metal_bands2)

Residuals:
    Min       1Q   Median       3Q      Max
-2.79612 -0.45529  0.04414  0.48408  1.60000

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.73622    0.57267   1.286  0.2016
log(bands_pm)  0.08171    0.03775   2.165  0.0328 *
log(gdp_pc_2021) 0.53624    0.06708   7.994 2.34e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6006 on 100 degrees of freedom
(19 observations deleted due to missingness)
Multiple R-squared:  0.6784,    Adjusted R-squared:  0.6719
F-statistic: 105.5 on 2 and 100 DF,  p-value: < 2.2e-16
```

A 10% increase of metal bands per million is associated with an increase in mean expected happiness of 0.008 after adjusting for GDP per capita.



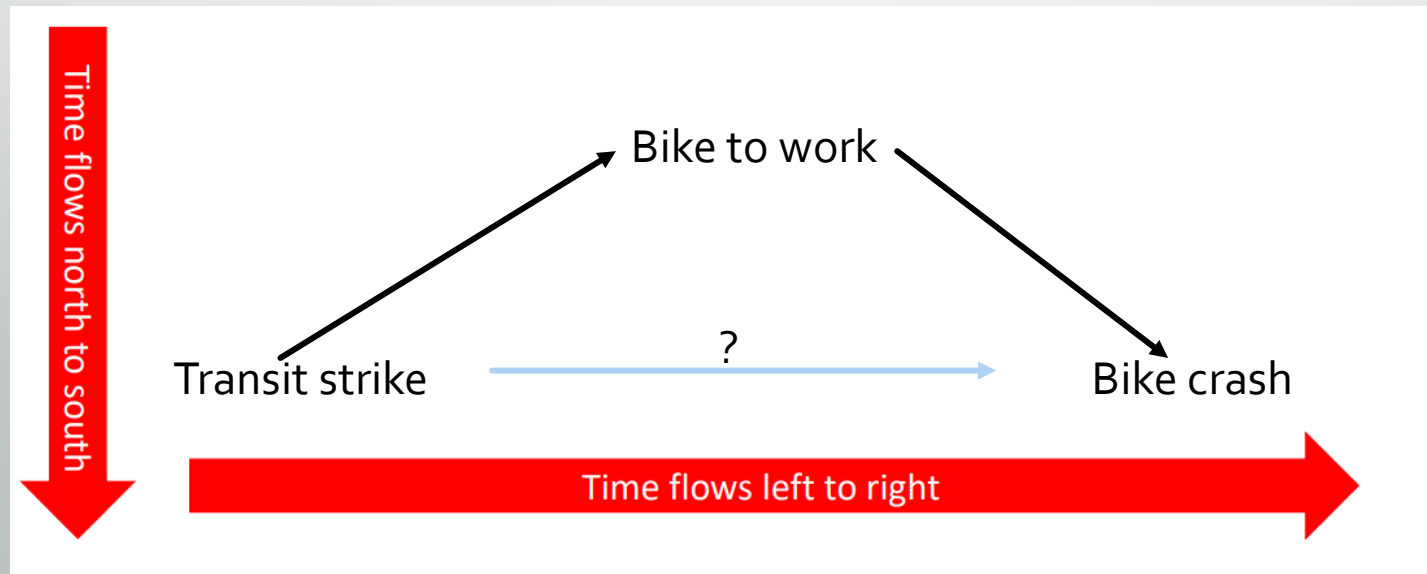
DAGs a very brief intro

Directed Acyclic Graphs (DAGs)

- DAGS = Directed Acyclic Graphs
- DAGs are a tool commonly used by researchers to conceptualize the relationship being modeled
- Directed: →
- Acyclic: 

Directed Acyclic Graphs (DAGs)

- The presence of an arrow indicates the possibility/assumption of a relationship, not that it absolutely exists
 - This assumption should be based on prior evidence!

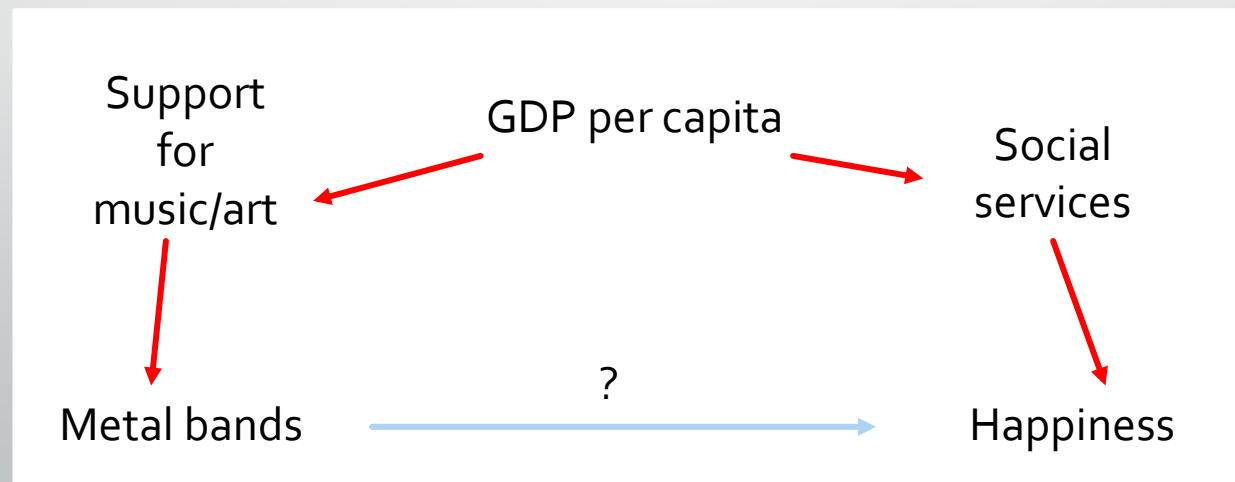


Directed Acyclic Graphs (DAGs)

- How might we represent the heavy metal/happiness model in a DAG?

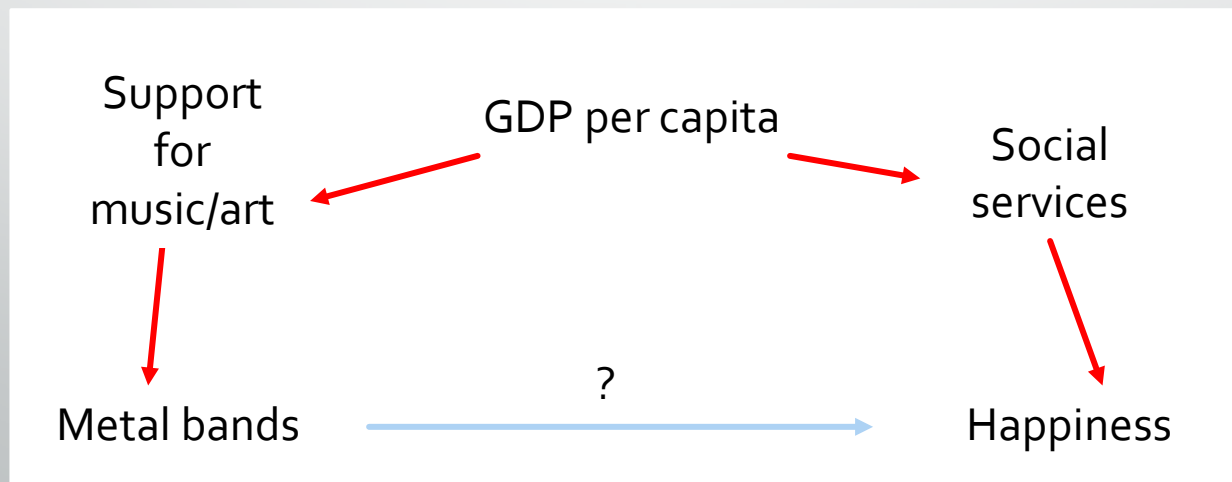
Directed Acyclic Graphs (DAGs)

- How might we represent the heavy metal/happiness model in a DAG?



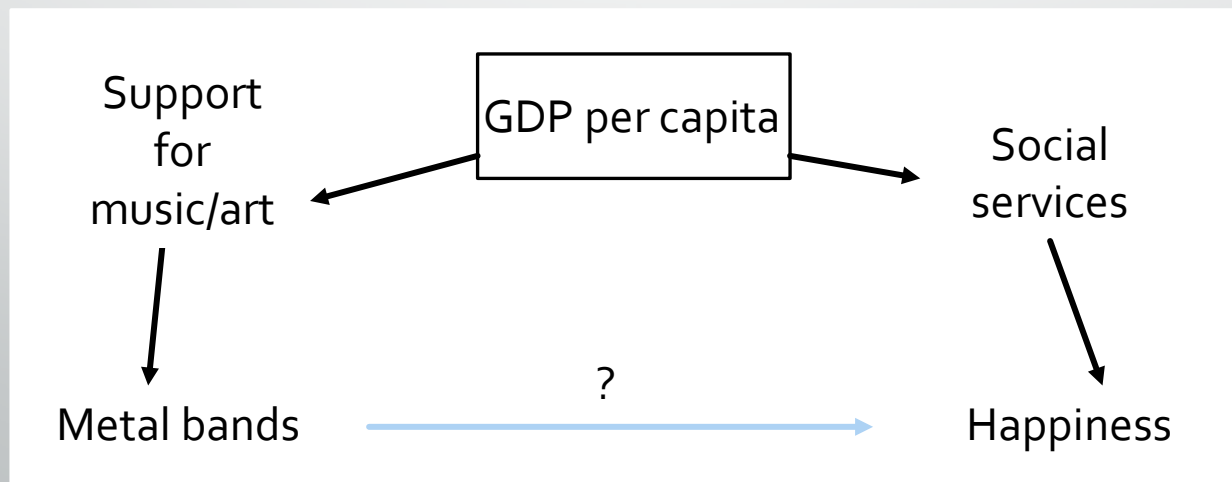
Directed Acyclic Graphs (DAGs)

- How might we represent the heavy metal/happiness model in a DAG?
 - Open “backdoor path” from the exposure (metal bands) through support for music/art, GDP per capita, and social services to the outcome (happiness).
 - What happens when we “control” for GDP per capita by including it as a predictor?



Directed Acyclic Graphs (DAGs)

- How might we represent the heavy metal/happiness model in a DAG?
 - Controlling for GDP per capita by including it in the model closes the backdoor path allowing us to assess whether the concentration of metal bands in a country is associated with the average happiness of its citizens.



We will talk more about DAGs in a few weeks.



Multiple Linear Regression

Multiple Linear Regression

Y (Income)	X_1 (Education)	X_2 (Senority)
99.917	21.586	113.103
92.579	18.276	119.310
34.679	12.069	100.690
$\vdots$	$\vdots$	$\vdots$
74.611	17.034	106.897

$$\begin{pmatrix} 99.917 \\ 92.579 \\ 34.679 \\ \vdots \\ 74.611 \end{pmatrix} = \beta_0 + \beta_1 \times \begin{pmatrix} 21.586 \\ 18.276 \\ 12.069 \\ \vdots \\ 17.034 \end{pmatrix} + \beta_2 \times \begin{pmatrix} 113.103 \\ 119.310 \\ 100.690 \\ \vdots \\ 106.897 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \vdots \\ \epsilon_N \end{pmatrix}$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

Multiple Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

$$Y = X\beta + \epsilon$$

- What does β_0 mean? Do we need β_0 ?
- What about β_1 and β_2 ? Do we need β_1 and β_2 ?
- What does ϵ mean? Do we need ϵ ?

Multiple Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

$$Y = X\beta + \epsilon$$

In multiple linear regression the intercept can be interpreted as the expected mean of y when all predictors equal 0 or their reference group.

- What does β_0 mean? Do we need β_0 ?
- What about β_1 and β_2 ? Do we need β_1 and β_2 ?
- What does ϵ mean? Do we need ϵ ?

Multiple Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

$$Y = X\beta + \epsilon$$

- What does β_0 mean? Do we need β_0 ?
- What about β_1 and β_2 ? Do we need β_1 and β_2 ?
- What does ϵ mean? Do we need ϵ ?

In multiple linear regression each beta coefficient represents the expected change in y for every one unit change in x while holding the other predictors constant.



Refresher from Simple Linear Regression

OLS Estimators of Intercept and Slope; Error variance

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \equiv \frac{SS_{XY}}{SS_X} = \frac{SS_{XY}(n-1)^{-1}}{SS_X(n-1)^{-1}} = \frac{s_{XY}}{s_{XX}}$$

$$\hat{\beta}_0 = \bar{Y} - \bar{X}\hat{\beta}_1$$

(From property 2)

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n \hat{\epsilon}_i^2}{n-2} = \frac{RSS}{n-2}, \text{ where}$$

$$\text{Residual: } \hat{\epsilon}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i,$$

$$\text{RSS(Residual Sum of Squares): } \sum_{i=1}^n \hat{\epsilon}_i^2 = SS_Y - \frac{SS_{XY}^2}{SS_X}$$

OLS Assumptions

Will see these assumptions presented a variety of ways (e.g., some are often combined), but they boil down to the following:

1. There is a linear relationship between the predictor(s) X and the outcome Y . – Relatively simple assumption to relax as we will see.
2. Variable, $X_1, \dots, X_n$, are fixed constants such that $S_{XX} > 0$
3. Variable, $\epsilon_1, \dots, \epsilon_n$, are random variables
4. Zero Error Mean: $E(\epsilon_i) = 0$ for $i = 1, \dots, n$
5. Uncorrelated errors: $\epsilon_1, \epsilon_2, \dots, \epsilon_n$ are not correlated – i.e., knowing value of ϵ_1 tells you nothing about $\epsilon_2, \dots, \epsilon_n$
6. Constant error variance: $V(\epsilon_i) = \sigma^2$ -- AKA homoscedasticity
7. Error normality: $\epsilon_i \sim N(0, \sigma^2)$ for $i = 1, \dots, n$

Properties of OLS

1. OLS estimators are unbiased as $E(\hat{\beta}_0, \hat{\beta}_1) = (\beta_0, \beta_1)$
2. Estimated error variance through residuals is unbiased as $E(\hat{\sigma}^2) = \sigma^2$
3. We can express variance-covariance of OLS estimates
4. We can derive unbiased estimators of OLS variance-covariance
5. OLS Estimators=Best Linear Unbiased Estimators
6. Regression coefficients follow a bivariate normal distribution
7. Estimated error variance, $\hat{\sigma}^2$, follows a χ^2 distribution



Multiple Linear Regression Estimation

OLS Estimation for Multiple Regression Model

$$\begin{aligned}\text{OLS minimizes } \sum_{i=1}^n \epsilon_i^2 &= \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2 \\ \Leftrightarrow \epsilon^T \epsilon &= (Y - X\beta)^T (Y - X\beta) \quad (\because Y = X\beta + \epsilon)\end{aligned}$$

Remember that for simple linear regression:

$$\text{OLS minimizes } \epsilon^T \epsilon = (Y - X\beta)^T (Y - X\beta)$$

We find $\hat{\beta}$ that satisfies $X^T X \hat{\beta} = X^T Y$

$$\Rightarrow \hat{\beta} = (X^T X)^{-1} X^T Y$$



Interpretation of Regression Coefficients

Interpretation of $\beta_j, j = 1, \dots, p$

Interpretation 1: Marginal effect of a unit change in the j^{th} predictor variable on the outcome variable while keeping other predictor variables constant.

Interpretation 2: Marginal effect of a unit change in the j th predictor variable on the outcome variable after removing the effect of all other predictor variables from the j th predictor variable.

Interpretation 3: Marginal effect of a unit change in the j th predictor variable on the outcome variable after removing the effect of all other predictor variables from both the j th predictor variable and the outcome variable.

Interpretation of $\beta_j, j = 1, \dots, p$

Interpretation 1: Marginal effect of a unit change in the j^{th} predictor variable on the outcome variable while keeping other predictor variables constant.

Interpretation 2: Marginal effect of a unit change in the j^{th} predictor variable on the outcome variable after removing the effect of all other predictor variables from the j^{th} predictor variable.

Interpretation 3: Marginal effect of a unit change in the j^{th} predictor variable on the outcome variable after removing the effect of all other predictor variables from both the j^{th} predictor variable and the outcome variable.

Interpretation of $\beta_j, j = 1, \dots, p$

Interpretation 1: Marginal effect of a unit change in the j^{th} predictor variable on the outcome variable, holding all other predictor variables constant.

Interpretation 2: Marginal effect of a unit change in the j^{th} predictor variable on the outcome variable, holding all other predictor variables constant. Put another way – With a numeric predictor it can be interpreted just as in a simple linear regression model and adding the statement “while holding other predictors constant”.

Interpretation 3: Marginal effect of a unit change in the j^{th} predictor variable on the outcome variable after removing the effect of all other predictor variables from both the j^{th} predictor variable and the outcome variable.



Inference about Regression Coefficients

Testing Predictors - 1

- How do we test predictors?

e.g., $H_0: \beta_1 = 0$

- How about a subset of predictors?

e.g., $H_0: \beta_1 = \beta_2 = 0$

- How about whether the effect of two predictors is the same?

e.g., $H_0: \beta_1 = \beta_2$

- How about a preset effect?

e.g., $H_0: \beta_1 = 2$

#2-4 in "Class5_Lab.rmd"

Testing Predictors - 2

Property 6 - $\hat{\beta} \sim N_p(\sigma^2(X^T X)^{-1})$

- Each $\hat{\beta}_j$ follows a normal distribution

For a single parameter, β_j :

- Studentized t-tests with: $t = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$
- Test $H_0: \beta_j = 0$
- $100(1 - \alpha)$ confidence interval of β : $\hat{\beta}_j \pm t_{n-p}^{\frac{\alpha}{2}} SE(\hat{\beta}_j)$

Testing predictors - 3

For multiple parameters, β :

- Test $H_0: \beta_1 = \beta_2 = \dots = \beta_p = 0$ and each $\beta_j = 0$

Bootstrap Confidence Intervals

Without **Assumption 5**. $\epsilon_i \sim N(0, \sigma^2)$ for $i = 1, \dots, n$

Bootstrap:

- Generate ϵ^* from $\hat{\epsilon}_1, \dots, \hat{\epsilon}_n$
- Generate $y^* = X\hat{\beta} + \epsilon^*$
- Estimate $\hat{\beta}^*$ by modeling y^* on X
- Repeat and observe $\hat{\beta}^*$

***We will come back to this in Class 14**



Analysis of Variance in Linear Regression

Analysis of Variance Table

Source	DF	Sum of Sq	Mean Sq	F
$\mathbf{X}$ (Model)	$p - 1$	SS_{Reg}	$MS_{Reg} = SS_{Reg}/(p - 1)$	MS_{Reg}/MSE
Error	$n - p$	RSS	$MSE = RSS/(n - p)$	
Total	$n - 1$	SS_Y		

- $SS_Y = \sum (Y_i - \bar{Y})^2 = \sum (Y_i - \hat{Y}_i + \hat{Y}_i - \bar{Y})^2 = RSS + SS_{Reg} \rightarrow SS_Y \geq RSS$
 - $RSS = \sum_{i=1}^n \hat{\epsilon}_i^2 = \sum (Y_i - \hat{Y}_i)^2 = SS_{YY} - SS_{Reg}$
 - $SS_{Reg} = \sum (\hat{Y}_i - \bar{Y})^2 = SS_Y - RSS$
 - $MSE = \hat{\sigma}^2 = \frac{(n-1)s_{\hat{\epsilon}}^2}{(n-p)} : E(MSE) = \sigma^2$ due to $\epsilon_i \sim N(0, \sigma^2)$
 - $F = \frac{(SS_Y - RSS)/(p-1)}{RSS/(n-p)} = \frac{MS_{Reg}}{MSE} \sim F_{n-p}^{p-1}$
- We can test $H_0: \beta_0 = \beta_1 = \dots = \beta_p = 0$ with F



Model Comparison

General F Test

- The more predictors in a model, the better the model “explains” the outcome variable in the data (unless $Y \perp\!\!\!\perp X$)
- However, **we want parsimony** in model
- How do we know which model is better? → General F Test

Consider a full model with p predictors (including an intercept) and a reduced model with q predictors.

$$\begin{aligned} F &= \frac{(RSS_{Reduced} - RSS_{Full}) / (df_{Reduced} - df_{Full})}{RSS_{Full} / df_{Full}} \\ &= \frac{(RSS_{Reduced} - RSS_{Full}) / (p - q)}{RSS_{Full} / (n - p)} \sim F_{n-p}^{p-q} \end{aligned}$$

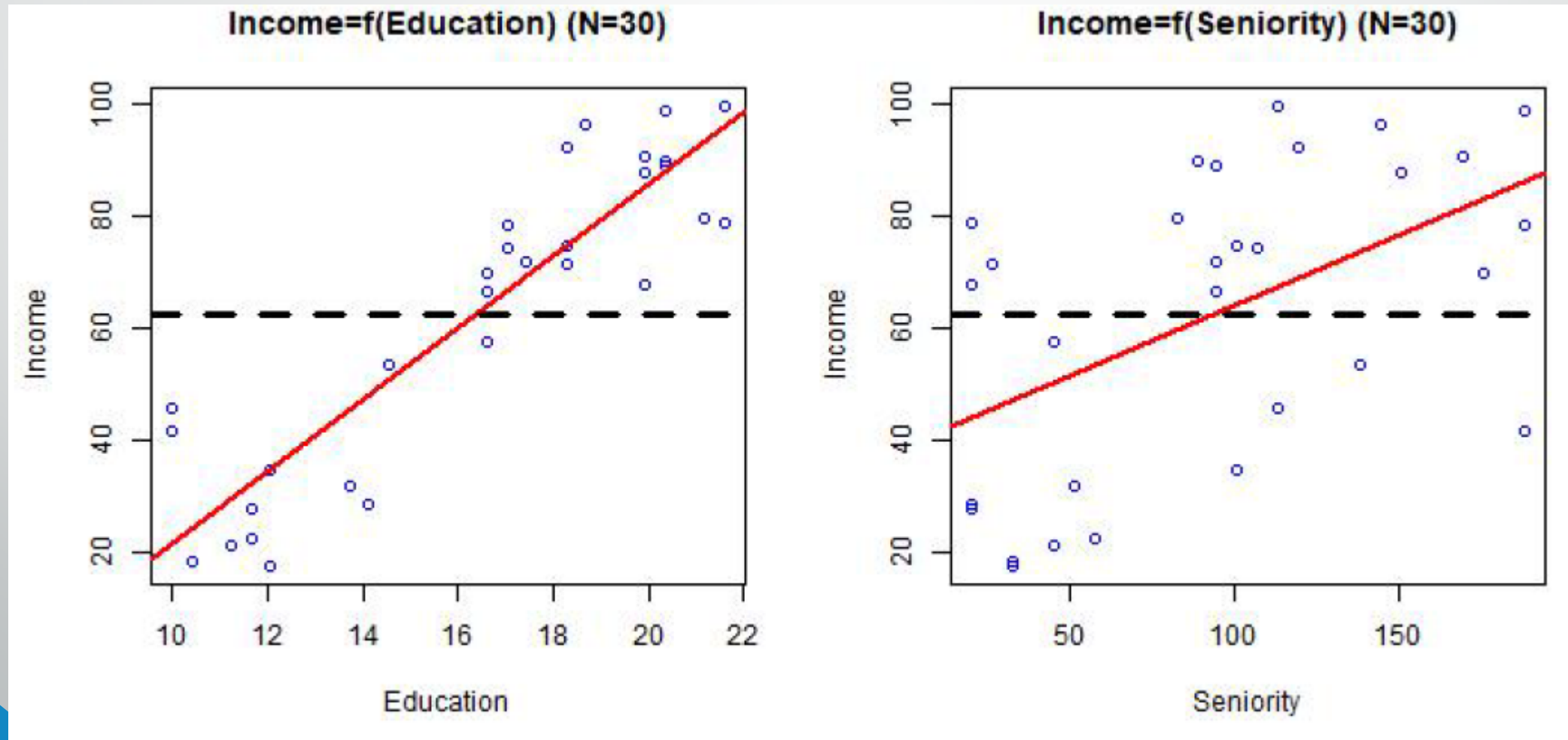
→ We can test $H_0 : RSS_{Reduced} = RSS_{Full}$ with F



Goodness of Model Fit

Model Fit

Which of the 3 lines explains the pattern in Income the best?



$$R^2 - 1$$

- $R^2 = \frac{SS_{Reg}}{SS_Y} = \frac{SS_Y - RSS}{SS_Y} = 1 - \frac{RSS}{SS_Y}$
- $0 < R^2 < 1$
- $F = \frac{SS_Y - RSS}{RSS} \frac{n-p}{p-1} = \frac{R^2}{1-R^2} \frac{n-p}{p-1}$

by using $\frac{SS_Y - RSS}{RSS} = \frac{SS_Y - RSS}{RSS} \frac{1}{RSS/SS_Y} = R^2 \frac{1}{RSS/SS_Y}$ and

$$\frac{RSS}{SS_Y} = \frac{SS_Y}{SS_Y} + \frac{RSS}{SS_Y} - \frac{SS_Y}{SS_Y} = 1 - \frac{SS_Y - RSS}{SS_Y} = 1 - R^2$$

- Think about R^2 for simple and multiple linear regression: is parsimony considered?

$$R^2 - 2$$

- Adjusted $R^2 = 1 - \frac{(1-R^2)(n-1)}{(n-p)}$
 - Considers parsimony by accounting for the sample size, n , and number of parameters, p
 - What does $n-p$ represent?

R^2 - 3

- Popular for use as goodness of fit measure
- Be careful for regression through the origin (e.g., model with no intercept) – will not be interpretable in the same way!
- Use R^2 in conjunction with the residual distribution (esp. $\hat{\sigma}^2$: MSE) for assessing model fit

Summing up

- Multiple linear regression provides us a valuable approach to explore relationships while accounting for other variables.
- This combined with prior knowledge and tools like DAGs we can be more intentional about the specific potential effect/relationship we're trying to model.
- Tools like the general F test, R^2 , and MSE can be useful in assessing whether one model fits the data better than another, but should be used carefully.
 - On their own, they can't tell us whether a causal effect exists.